# Alessandra Simmons

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## Education

**Northeastern University**, Boston, MA — Sep 2022 - Present

*BS in Computer Science and Philosophy, Minors in Computational Social Science and Sociology* — Expected: May 2026

GPA: 3.79/4.0

Coursework: Theory of Computation, Advanced Logic, Logic and Computation, Computer Networks, Computer Systems, System Security, Software Engineering (IP), Algorithms (IP), Object-Oriented Design, Intro to CS Research

## Skills

**Operating Systems**: Linux (Fedora, Debian, Arch), Windows

**Programming Languages**: Rust, Lean 4, C, C++, Java, Scala, Verilog, Python, Scheme/Racket

**Areas of Experience**: Formal verification, Compilers, Embedded systems, Theorem proving, Hardware design

## Experience

**Software Development Engineer I** · AMD Inc. · Boxborough, MA — Jun 2026 - Future (planned)

- Working on Comgr, the object management tool for AMD's GPU compiler toolchain
- Making upstream contributions to LLVM

**Compiler Development Intern** · AMD Inc. · Boxborough, MA — Sep 2025 - Dec 2025

- Added support for `llvm-offload-packages` to Comgr which will enable the HIP runtime to be brought in line with planned LLVM defaults
- Contributed to LLVM to improve developer documentation and reduce tech debt

**Research Assistant** · Northeastern University · Boston, MA — Jan 2025 - Aug 2025

- Assisted with the design of an experimental, highly-parallel abstract computer architecture design with Professor Gunar Schirner and the Northeastern Novel Computer Architecture Group
- Created an MLIR compiler backend which targets a simulator of that computer

## Projects

**Indi** / *Lean 4, Formal methods, Theorem proving, React, Typescript* — Jan 2026 - Present

- Writing a complex rules engine for playing interactive *Magic: The Gathering*-compatible games in **Lean 4**, conformant to the specification laid out in the MtG Comprehensive Rules
- Writing a web app for delivering games from the rules engine in Typescript and React

**Aardvark Compiler** / *Compilers, MLIR, Rust, Formal methods, RISC-V* — April 2025 - Nov 2025

- Created a compiler frontend in **Rust** which converts a pure and functional ML-like language into **MLIR** using a hand-rolled parser, typechecker, and compilation passes, verified using property-based testing
- Wrote a compiler backend which converts the MLIR dialects of the frontend into a **RISC-V** Linux ELF

**Juniper CAS** / *Formal methods, Lean 4, Rust, Computer algebra* — Nov 2024 - Feb 2025

- Used **Lean** metaprogramming to automatically elaborate and convert theorem types into portable data
- Created a formally-specified Computer Algebra System in **Rust** by automatically converting those Lean theorems into Rust rewriting rules for doing equality saturation with the Egg e-graph library

**Formalizing Game Theory** / *Formal methods, Lean 4, Game Theory, Theorem proving* — Dec 2023 - Feb 2025

- Devised a formalization of strategic game theory in the **Lean 4** proof language, allowing one to prove certain properties about specific games and strategies and about generalizations of games and strategies
- Wrote tools for game-related **proof automation**, along with examples of formalized games and proofs

## Interests

Reading Brandon Sanderson and David Graeber · Playing Bass, Piano, and Guitar · Replaying Pokémon in Spanish